

Project 6 Elevator Graphical User Interface (GUI)

Introduction:

The Elevator Project requires both local hardware and remote software control of the physical elevator system. While both control methods require several hardware and software subsystems implemented in the elevator to work together to handle user requests, we will provide two different routes of entry for user requests. The local hardware method will interact with the supervisory controller (the Raspberry Pi) through the onboard CAN network and the floor controllers and car controller included in the elevator system. The remote software method will communicate with the supervisory controller over a locally-hosted server to implement the user request on the greater elevator system. The GUI will act to mimic and function in parallel with the floor controllers and car controller.

Equipment Description:

The elevator system is comprised of four main hardware groups: the supervisory controller, the elevator controller, the car controller, and the floor controllers. Brief details regarding each of these follow:

Supervisory Controller:

The supervisory controller is the main computational element in the elevator system; it manages the system's overall function. The supervisory controller:

- Implements a finite state machine to direct the elevator car to a desired floor based on the system's current state and a user request.
- Receives input from the car controller and floor controllers when a hardware request is given, and instructs the elevator controller how to articulate the elevator winch.
- Receives input from the elevator controller reporting the car's current position in the system.
- Dispatch the elevator car via the elevator controller, informed by local and remote requests.
- Communicate with the remote control system (GUI) using a locally-hosted MySQL database, where it will record information about hardware request, current system state, and request service completion.

Elevator controller:

The elevator controller will:

- Receive instructions from the supervisory controller regarding floor requests originating from the floor controllers, car controller, and the GUI.
- Broadcast the car's current position over the CAN network to all available nodes.

The elevator controller will not:

- Communicate directly with the GUI.

Car Controller:

The car controller will:

- Designate a floor that the elevator car should be sent to by interpreting controller-local button presses.
- Display the selected floor using an LED indicator.
- Send requests to the supervisory controller.

Floor Controllers:

The floor controllers will:

- Send requests to the supervisory controller to ask that the elevator car be brought to its floor.
- Display that a request has been submitted and is being served using an LED.

Graphical User Interface (GUI):

The graphical user interface is an alternate, remote control method for the elevator system which functions in parallel with the local request system (the combination of the floor controllers and the car controller). The supervisory controller manages competing requests between the local and remote request systems, prioritizing local requests. The GUI will:

- communicate with the supervisory controller through the MySQL database.
- Both send inputs and receive feedback from the database.
- Display car movement and service status using on-screen indicators.

GUI Design Decisions:

Since the GUI is a remote application, hosted on the project website, it is assumed that the user will not have access to the local request system or indicators, so the GUI should reproduce the function of the local request system. This means that the GUI should offer both car controller input options and indicators, and floor controller input options and indicators. The GUI will allow a user to select a chosen floor request, having the car sent to that floor as if the indicated floor's hardware call button had been pressed. It should display an indication that the request was received, by way of a visual change to the GUI, lighting the selected floor's button in the time between the request submission and the service receipt through the database. The user will subsequently be able to request another floor to where the car should be sent. The GUI will also indicate the car's direction of travel, and allow users to select an adjacent floor using the direction buttons. The GUI will monitor the physical state of the elevator system through the database, which is enabled by the supervisory controller's data-logging.